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Experiment No. 09

Aim : To provision cloud infrastructure using Terraform.

LO Mapped : LO4

LO : Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

- **IAM User / Account ID:** Experiment7 / 702561770822
- **Target AWS Region:** ap-southeast-2 (Asia Pacific - Sydney)
- **Deployment Status:** Successfully Provisioned, Verified, and Deprovisioned

1. Aim / Objective

To design, write, validate, and execute Infrastructure as Code (IaC) configurations using

HashiCorp Terraform to dynamically provision, configure, verify, and destroy an Amazon Elastic Compute Cloud (EC2) virtual machine and associated resources on Amazon Web Services (AWS).

2. Theoretical Background

What is Infrastructure as Code (IaC)?

Infrastructure as Code (IaC) is an IT infrastructure provisioning process where developers and DevOps engineers automatically manage and provision cloud resources (networks, virtual machines, load balancers, and storage) through human-readable definition files rather than manual user interface configurations. IaC enforces version control, repeatability, scalability, and consistency across environments.

Key Terraform Concepts & Lifecycle Commands

HashiCorp Terraform is an open-source IaC tool that uses HashiCorp Configuration Language (HCL) to define cloud infrastructure declarative state files.

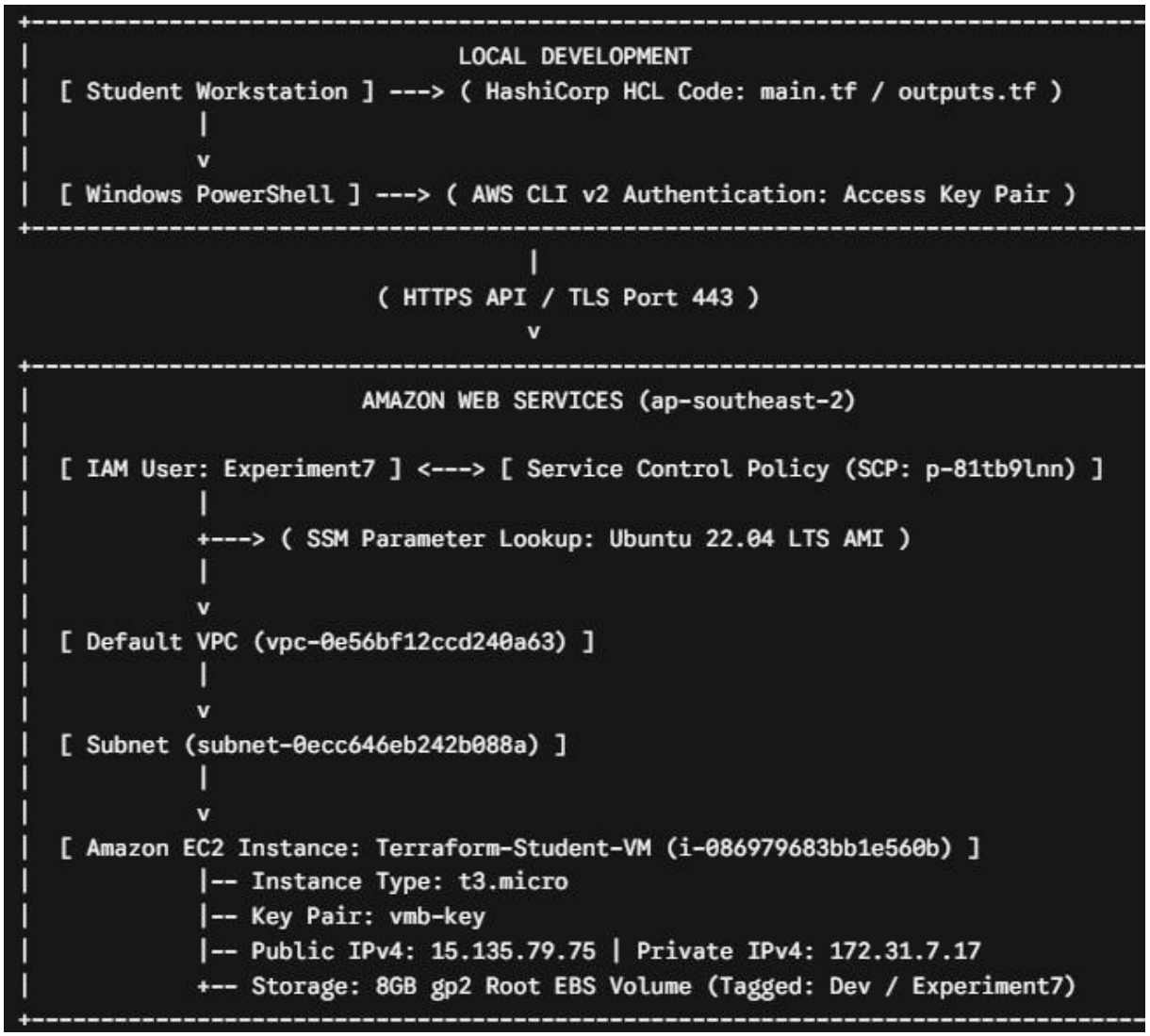
- **Declarative Configuration (.tf):** Files where target cloud resources are defined. All .tf files in a single execution directory are parsed together as a unified configuration.
- **terraform init:** Initializes the working directory, downloads required provider plugins (e.g., hashicorp/aws v6.67.0), and creates dependency lock files (.terraform.lock.hcl).
- **terraform validate:** Checks the syntax, structure, and internal consistency of HCL configuration files.
- **terraform plan:** Compares the local declarative configuration against the active cloud infrastructure state and constructs an execution plan without applying changes.
- **terraform apply:** Executes the proposed plan against AWS APIs to provision or modify cloud resources, saving state metadata into terraform.tfstate.
- **terraform output:** Extracts and displays defined output variables from the state file.
- **terraform destroy:** Deprovisions and cleans up all managed resources from AWS to prevent unnecessary billing.

Interfacing AWS Concepts & Security

1. **AWS Identity and Access Management (IAM):** Grants precise execution policies (AmazonEC2FullAccess, AmazonVPCFullAccess, AmazonSSMReadOnlyAccess) and security credentials (Access Key ID & Secret Access Key) for programmatic CLI access.
2. **AWS Systems Manager (SSM) Parameter Store:** Dynamically retrieves system parameters such as up-to-date, region-specific Canonical Ubuntu AMI IDs (/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/amiid).

3. **AWS Service Control Policies (SCPs):** Guardrails enforced at the AWS Organization level (p-81tb9lnn) that mandate resource compliance, such as explicit resource/volume tags (Environment = "Dev", Project = "Experiment7"), allowed instance types (t3.micro), and attached SSH Key Pairs (vmb-key).

3. Architecture Diagram & Workflow



4. Declarative Configuration Code

The configuration code was organized inside C:\Terraform.

1. main.tf (Infrastructure Definition)

```
Terraform terraform {  required_providers
{    aws = {      source  = "hashicorp/aws"
version = "~> 6.0"
   }
 }
}

provider "aws" {    region =
"ap-southeast-2"
}

# Dynamic SSM Parameter Lookup for Ubuntu 22.04 LTS AMI data "aws_ssm_parameter"
"ubuntu_ami" {  name          =
"/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebsgp2/ami-id"
}

# Default VPC and Subnet Fetching
data "aws_vpc" "default" {   default =
true
}

data "aws_subnets" "default" {
```

```
  filter {     name   = "vpc-id"     values =
[data.aws_vpc.default.id]
  }
}

# EC2 Instance Resource Definition resource "aws_instance"
"student_vm"    {            ami                                =
data.aws_ssm_parameter.ubuntu_ami.value   instance_type =
"t3.micro"    subnet_id       = data.aws_subnets.default.ids[0]
key_name      = "vmb-key"

  tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }

  volume_tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
}
```

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = "ap-southeast-2"
}

data "aws_ssm_parameter" "ubuntu_ami" {
  name = "/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id"
}

data "aws_vpc" "default" {
  default = true
}

data "aws_subnets" "default" {
  filter {
    name   = "vpc-id"
    values = [data.aws_vpc.default.id]
  }
}

resource "aws_instance" "student_vm" {
  ami           = data.aws_ssm_parameter.ubuntu_ami.value
  instance_type = "t3.micro"
  subnet_id     = data.aws_subnets.default.ids[0]
  key_name      = "vmb-key"

  tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
```

```
  volume_tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
}
```

2. outputs.tf (Output State Extraction) Terraform

output "instance_id" { description = "ID of the

created EC2 instance" value =

aws_instance.student_vm.id

}

output "instance_type" { description = "Instance type of

the created VM" value =

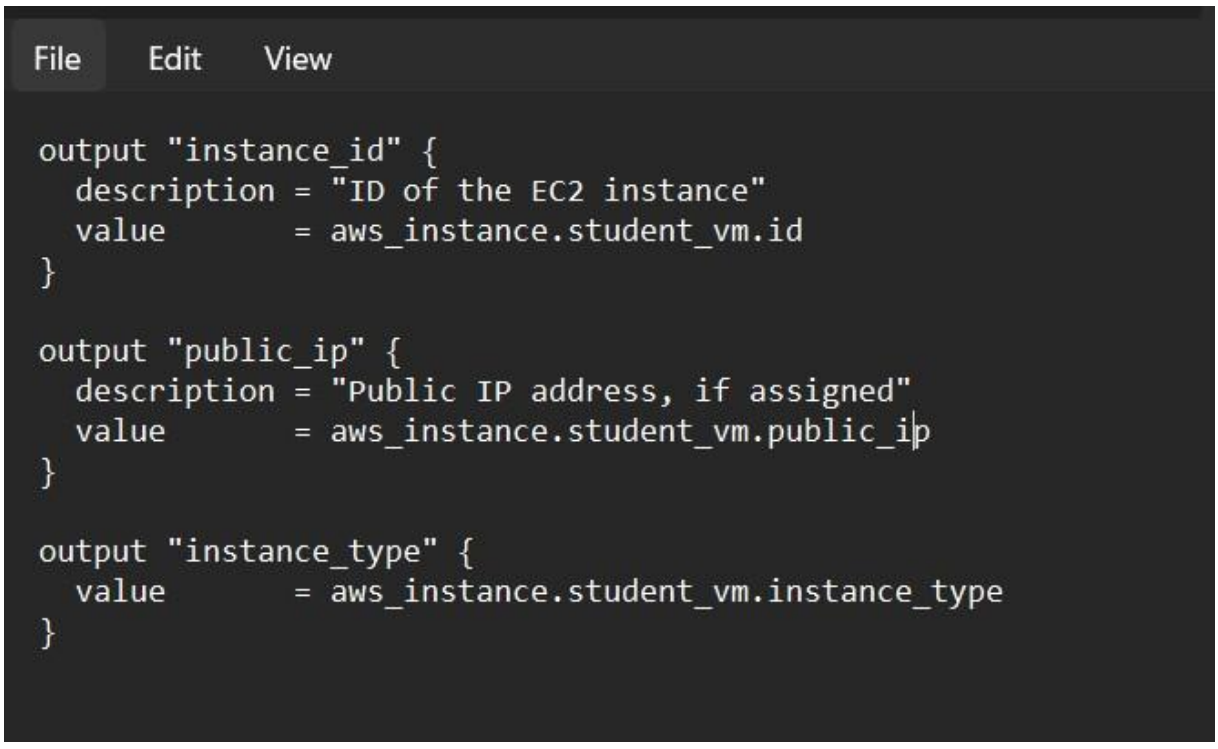
aws_instance.student_vm.instance_type

}

output "public_ip" { description = "Public IP address assigned to

the EC2 instance" value = aws_instance.student_vm.public_ip

}



5. Step-by-Step Implementation Procedure

Phase 1: Local Environment & Credentials Configuration

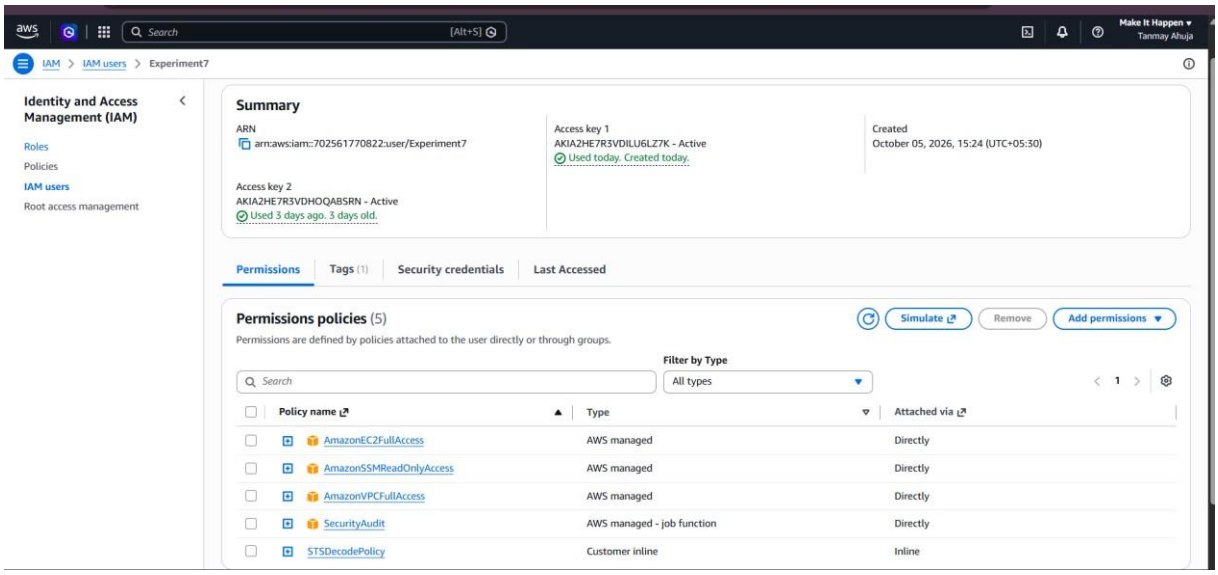
Step 1: AWS IAM Credentials & Access Key Setup

1. Log in to the AWS Management Console and navigate to **IAM > Users > Experiment7**.

2. Select **Security credentials**, click **Create access key**, and choose **Command Line Interface (CLI)**.

3. Generate and record the credentials:

   - **Access Key ID:** AKIA2HE7R3VDILU6LZ7K
   - **Secret Access Key:** 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



Step 2: AWS CLI Authentication & Verification

1. Open Windows PowerShell and run aws configure:

PowerShell

AWS Access Key ID: AKIA2HE7R3VDILU6LZ7K

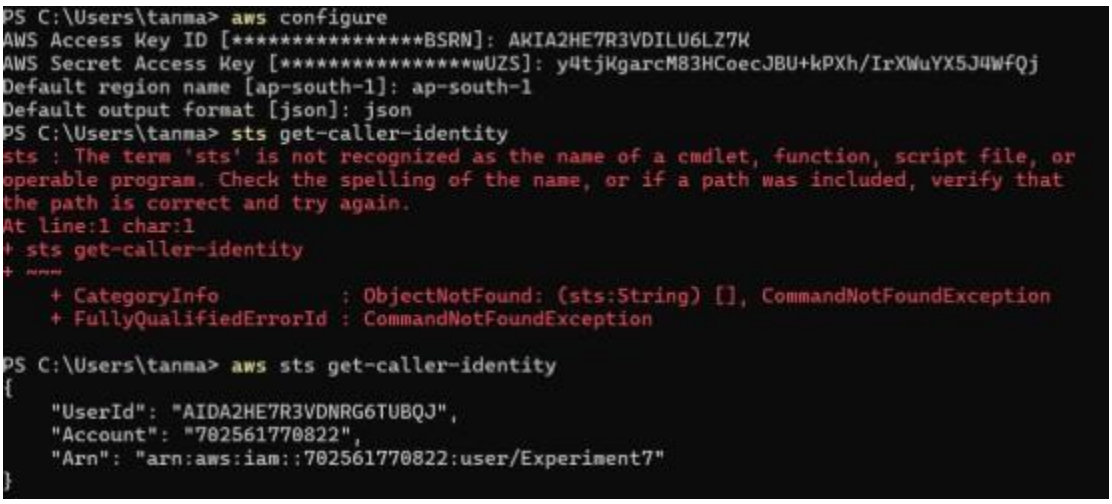
AWS Secret Access Key: y4tjKgarcM83HCoecJBU+kPXh/IrXWuYX5J4WfQj

Default region name: ap-southeast-2

Default output format: json

2. Verify identity via STS: PowerShell aws sts get-caller-

identity



.

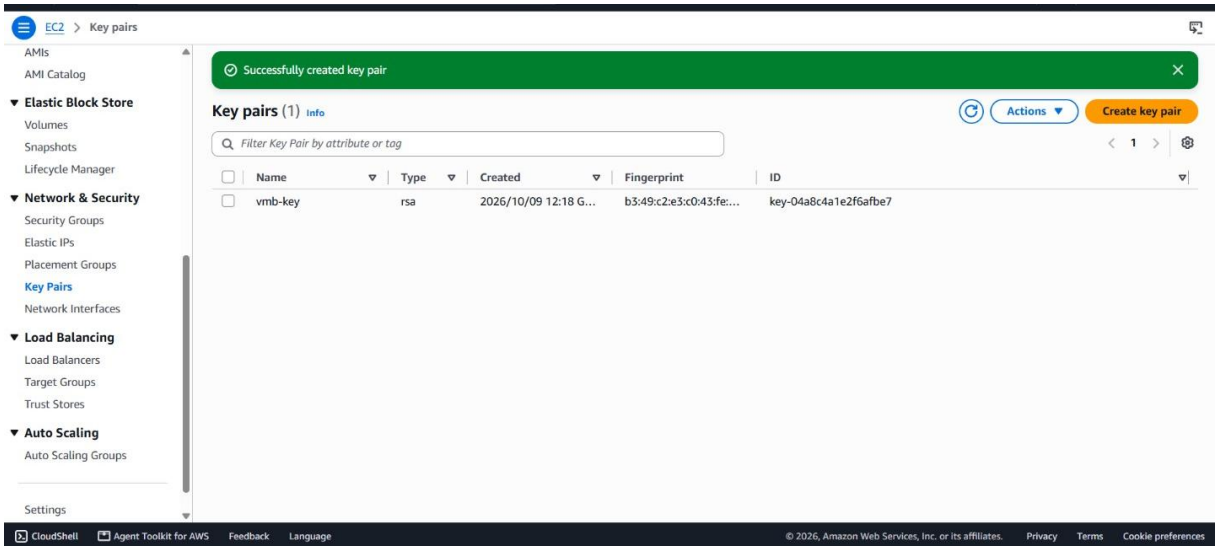
Step 3: EC2 Key Pair Generation in ap-southeast-2

1. Navigate to **AWS Console > EC2 > Key Pairs** in region ap-southeast-2.

2. Click **Create key pair**: ○ **Name:** vmb-key ○ **Key**

pair type: RSA ○ **Private key file format:** .pem 3.

Download and save vmb-key.pem.

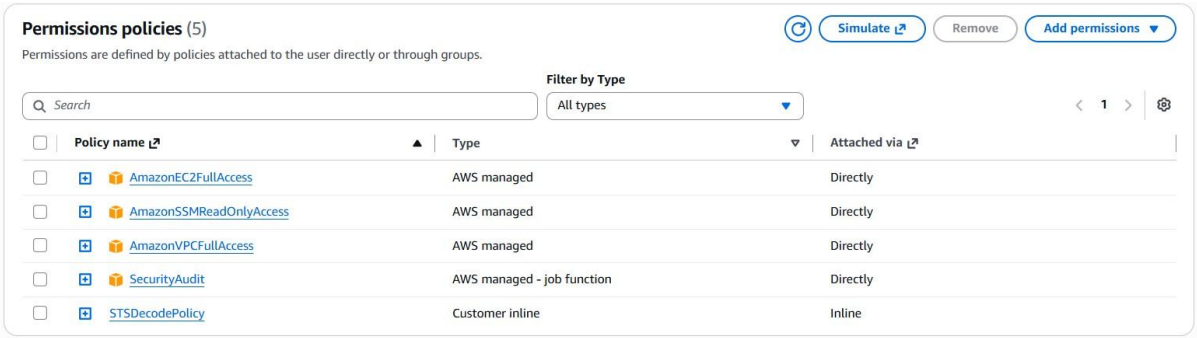


Phase 2: Resolving IAM Permissions & Policy Extensions

Step 4: Attaching SSM Read-Only Permissions

To enable Terraform to query the Canonical Ubuntu AMI ID dynamically from AWS Parameter Store, attach the SSM policy:

1. Go to **IAM > Users > Experiment7 > Permissions**.
2. Click **Add permissions > Attach policies directly**.
3. Search and select **AmazonSSMReadOnlyAccess**.
4. Click **Add permissions**.



Phase 3: Terraform Execution Lifecycle

Step 5: Directory Setup & Terraform Initialization

1. Create directory C:\Terraform and save main.tf and outputs.tf inside.

2. Open PowerShell, navigate to directory, and initialize backend: PowerShell

cd C:\Terraform terraform init

```
PS C:\Terraform> terraform init
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.67.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

Step 6: Code Validation & Execution Planning

1. Validate syntax:

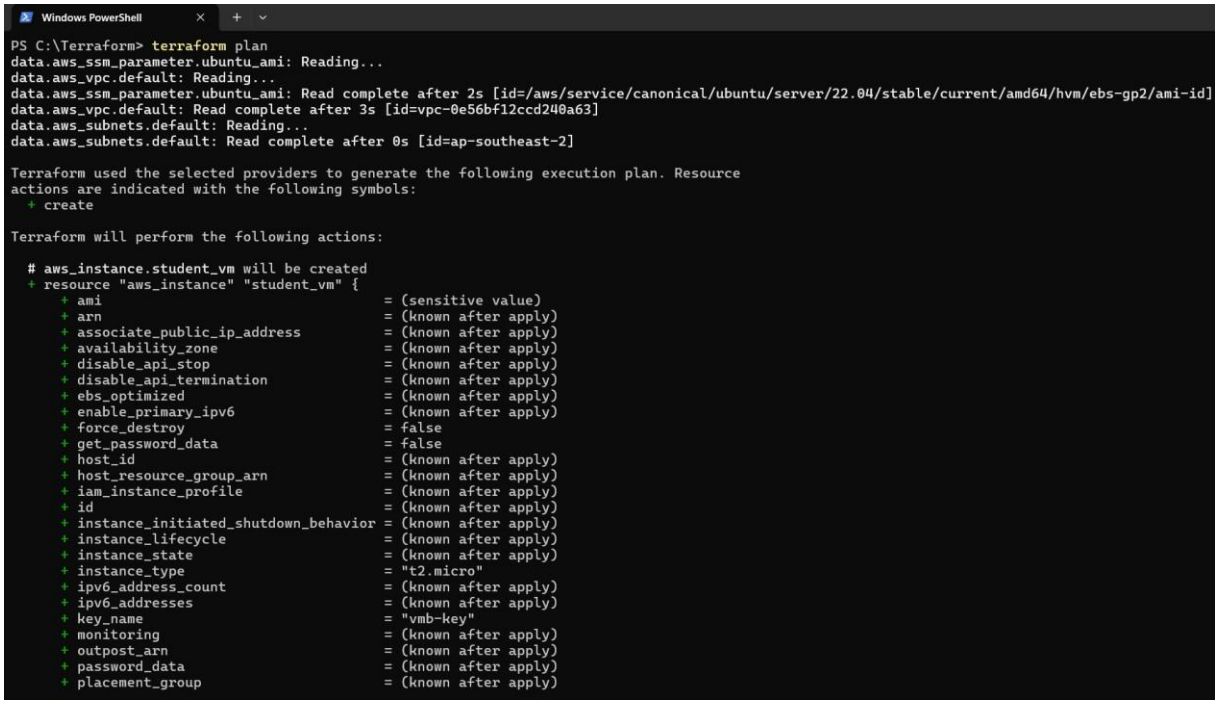
PowerShell terraform validate

2. Generate execution

plan: PowerShell terraform

plan

```
PS C:\Terraform> terraform validate
Success! The configuration is valid.
```

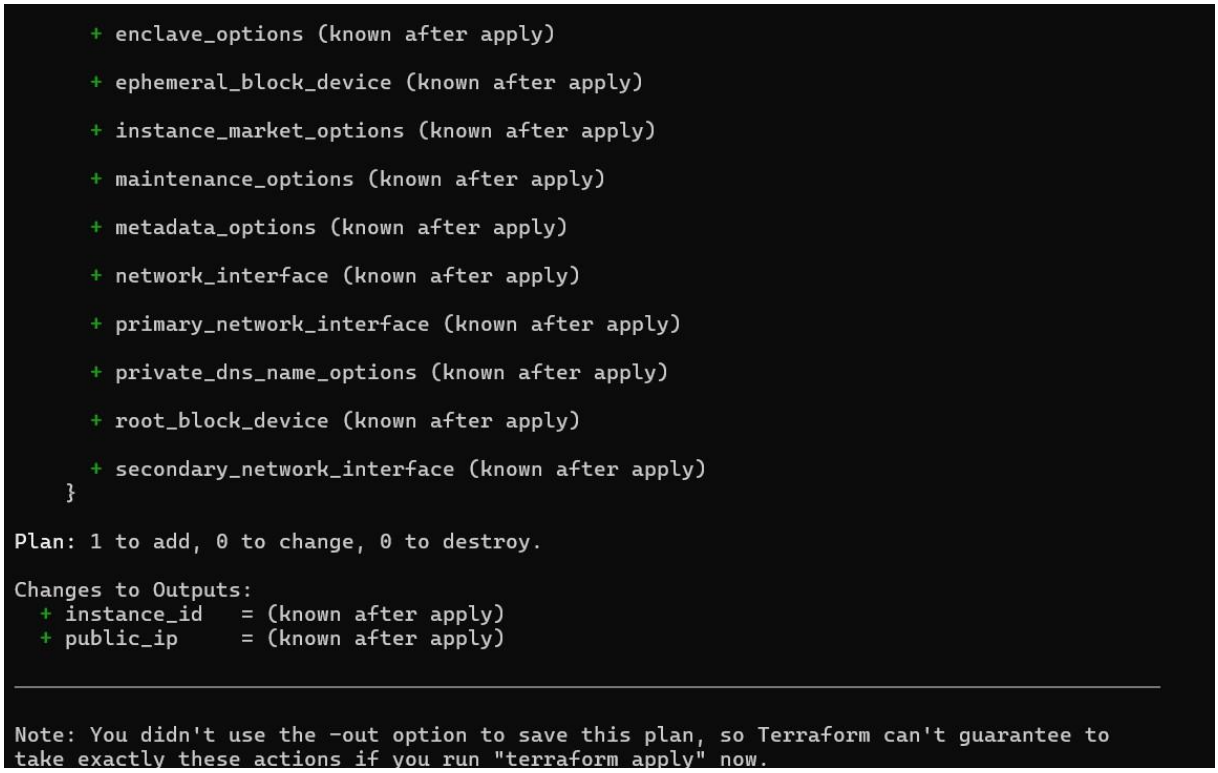


```
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-southeast-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = "subnet-0ecc646eb242b088a"
      + tags                                 = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tags_all                             = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + volume_tags                          = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)
```



Step 7: Applying Infrastructure Provisioning

1. Execute apply command: PowerShell terraform apply
2. Enter **yes** at the approval prompt.
3. Confirm resource creation completion.

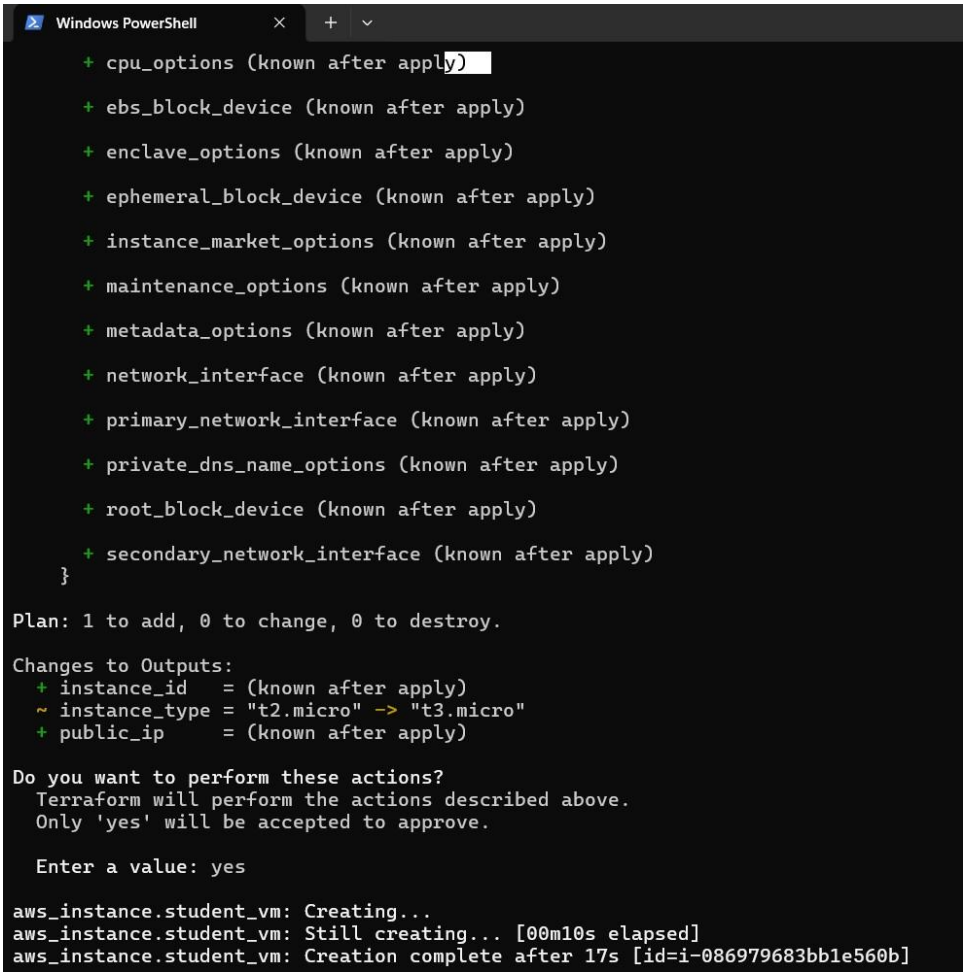
```
PS C:\Terraform> terraform apply
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 2s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 1s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.student_vm will be created
  + resource "aws_instance" "student_vm" {
      + ami                                  = (sensitive value)
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "vmb-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
```


Windows PowerShell
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_group_id = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns = (known after apply)
+ private_ip = (known after apply)
+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ region = "ap-southeast-2"
+ secondary_private_ips = (known after apply)
+ security_groups = (known after apply)
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = "subnet-0ecc646eb242b088a"
+ tags = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ tags_all = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ tenancy = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ volume_tags = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)




```
Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_id   = (known after apply)
  ~ instance_type = "t2.micro" -> "t3.micro"
  + public_ip     = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.student_vm: Creating...
aws_instance.student_vm: Still creating... [00m10s elapsed]
aws_instance.student_vm: Creation complete after 17s [id=i-086979683bb1e560b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

instance_id = "i-086979683bb1e560b"
instance_type = "t3.micro"
public_ip = "15.135.79.75"
```

Phase 4: Verification & Teardown

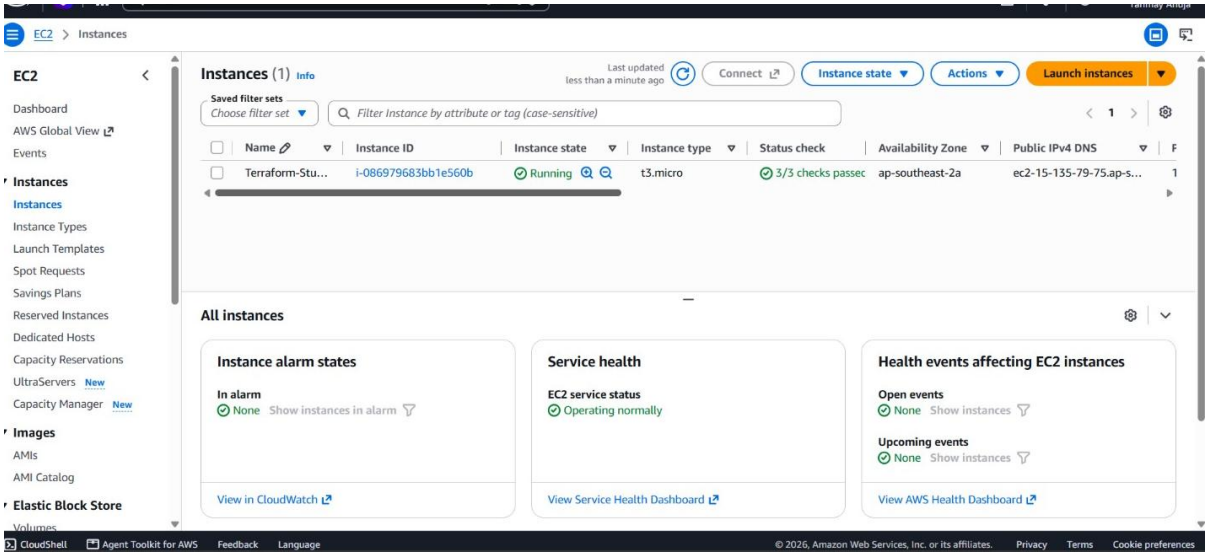
Step 8: Verifying Terraform Outputs & AWS EC2 Console State

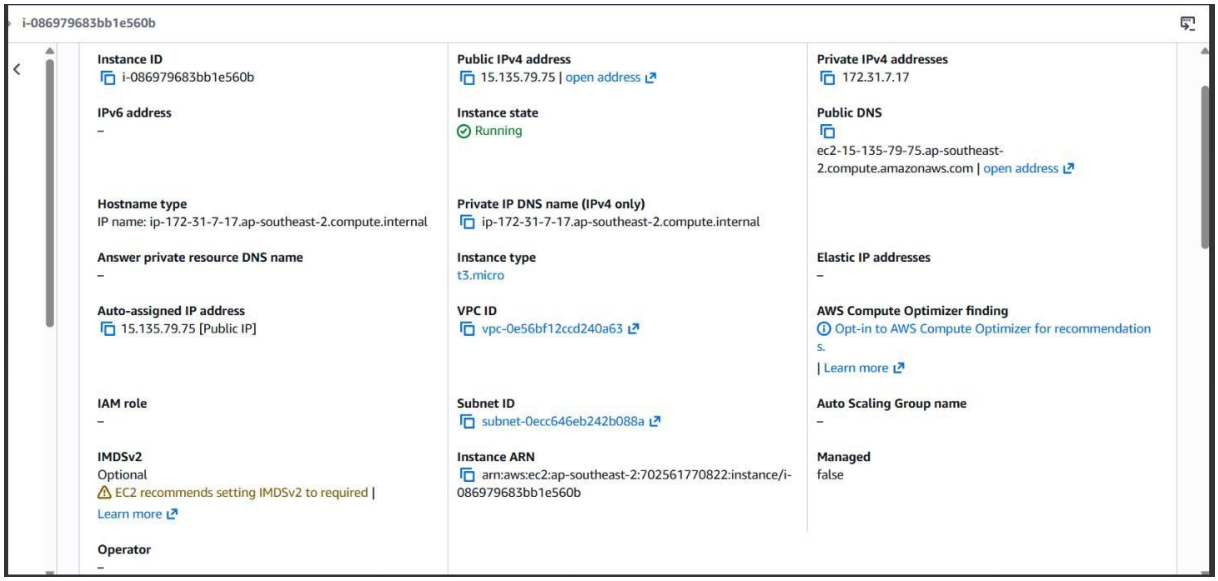
1. Run output command in PowerShell: PowerShell terraform output

2. Open browser and navigate to **AWS EC2 Dashboard > Instances** in region apsoutheast-2.

3. Verify that Terraform-Student-VM (i-086979683bb1e560b) is running with public IP 15.135.79.75.

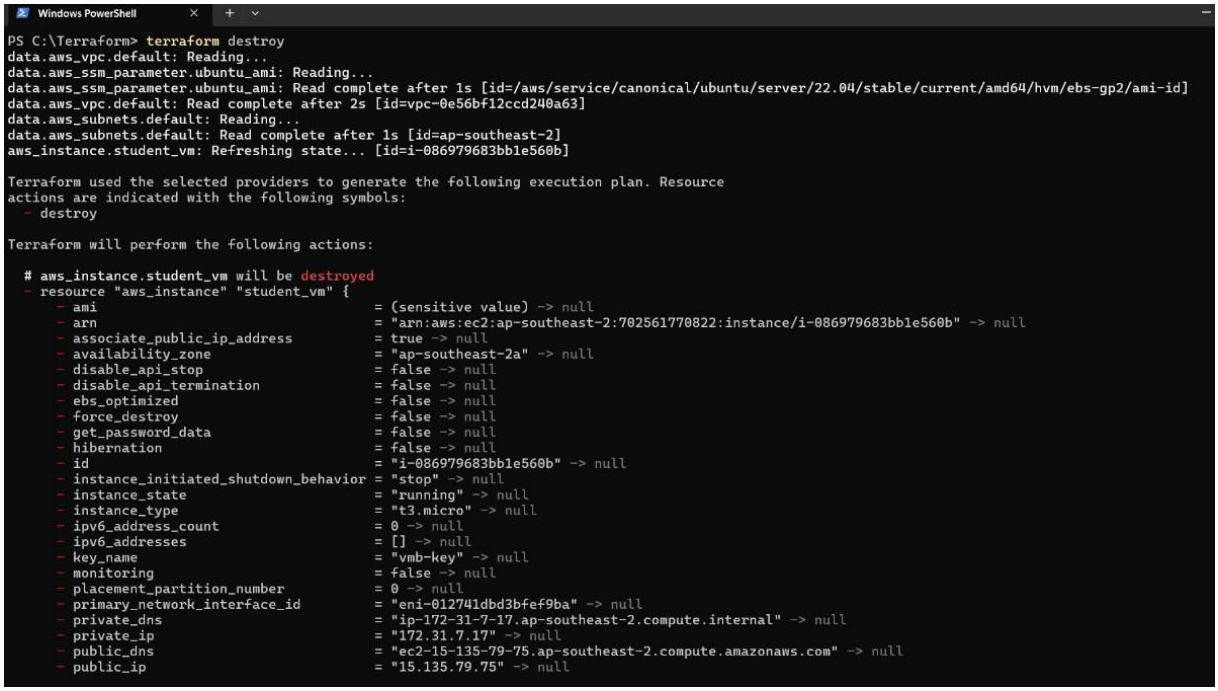
```
PS C:\Terraform> terraform output
instance_id = "i-086979683bb1e560b"
instance_type = "t3.micro"
public_ip = "15.135.79.75"
```





Step 9: Deprovisioning Cloud Resources (terraform destroy)

1. In PowerShell, initiate teardown: PowerShell terraform destroy
2. Type **yes** when prompted.
3. Confirm resource destruction.



```
Windows PowerShell

PS C:\Terraform> terraform destroy
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 2s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 1s [id=ap-southeast-2]
aws_instance.student_vm: Refreshing state... [id=i-086979683bb1e560b]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.student_vm will be destroyed
  - resource "aws_instance" "student_vm" {
      - ami                                  = (sensitive value) -> null
      - arn                                  = "arn:aws:ec2:ap-southeast-2:702561770822:instance/i-086979683bb1e560b" -> null
      - associate_public_ip_address          = true -> null
      - availability_zone                    = "ap-southeast-2a" -> null
      - disable_api_stop                     = false -> null
      - disable_api_termination              = false -> null
      - ebs_optimized                        = false -> null
      - force_destroy                        = false -> null
      - get_password_data                    = false -> null
      - hibernation                          = false -> null
      - id                                   = "i-086979683bb1e560b" -> null
      - instance_initiated_shutdown_behavior = "stop" -> null
      - instance_state                       = "running" -> null
      - instance_type                        = "t3.micro" -> null
      - ipv6_address_count                   = 0 -> null
      - ipv6_addresses                       = [] -> null
      - key_name                             = "vmb-key" -> null
      - monitoring                           = false -> null
      - placement_partition_number           = 0 -> null
      - primary_network_interface_id         = "eni-012741dbd3bfef9ba" -> null
      - private_dns                          = "ip-172-31-7-17.ap-southeast-2.compute.internal" -> null
      - private_ip                           = "172.31.7.17" -> null
      - public_dns                           = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
      - public_ip                            = "15.135.79.75" -> null
```



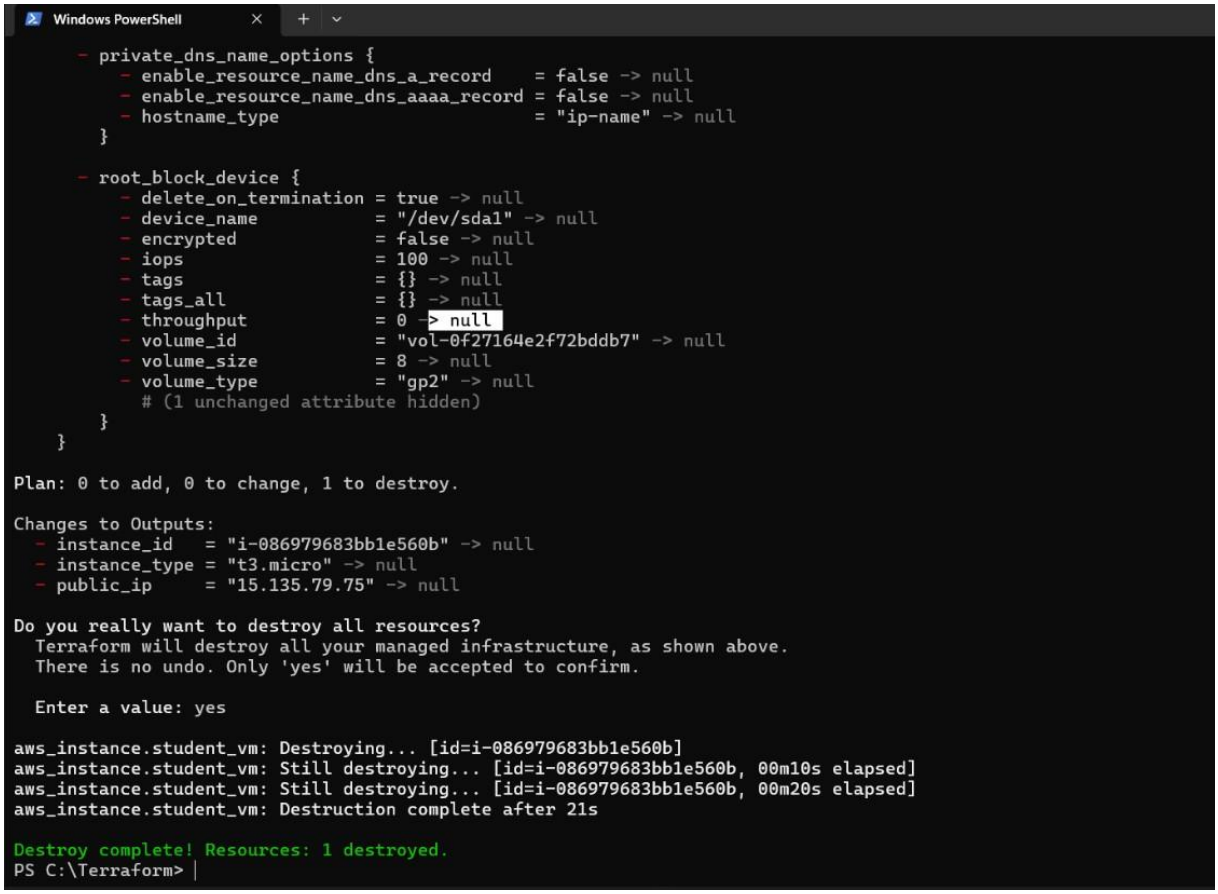
```
Windows PowerShell

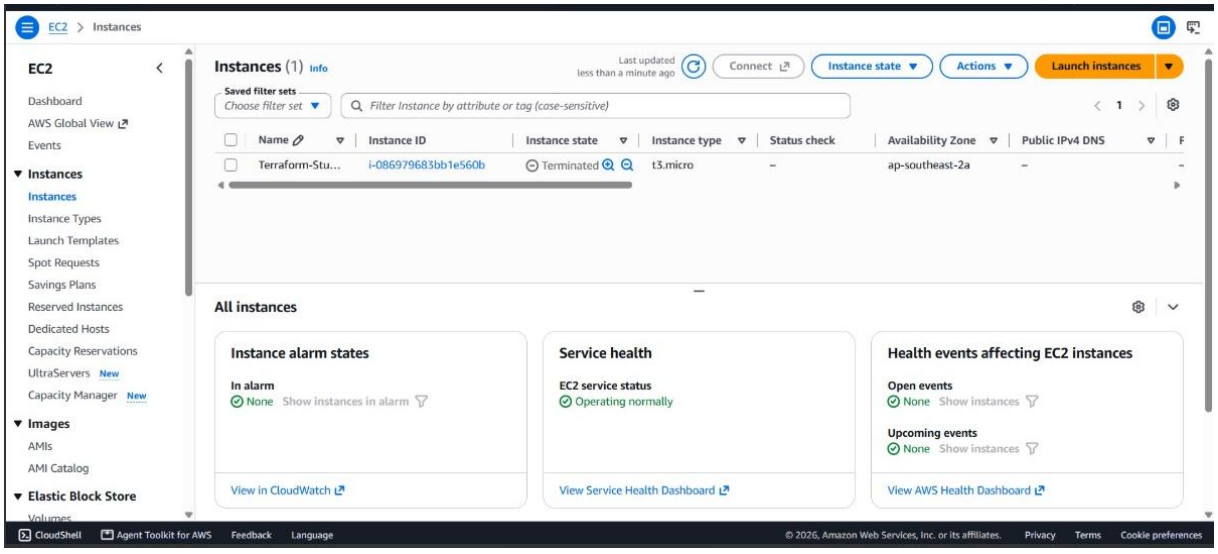
    - public_dns                            = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
    - public_ip                             = "15.135.79.75" -> null
    - region                                = "ap-southeast-2" -> null
    - secondary_private_ips                 = [] -> null
    - security_groups                       = [
        - "default",
      ] -> null
    - source_dest_check                     = true -> null
    - subnet_id                             = "subnet-0ecc646eb242b088a" -> null
    - tags                                  = {
        - "Environment" = "Dev"
        - "Name"        = "Terraform-Student-VM"
        - "Project"     = "Experiment7"
      } -> null
    - tags_all                              = {
        - "Environment" = "Dev"
        - "Name"        = "Terraform-Student-VM"
        - "Project"     = "Experiment7"
      } -> null
    - tenancy                               = "default" -> null
    - user_data_replace_on_change           = false -> null
    - volume_tags                           = {
        - "Environment" = "Dev"
        - "Name"        = "Terraform-Student-VM"
        - "Project"     = "Experiment7"
      } -> null
    - vpc_security_group_ids                = [
        - "sg-0b3493b246c88ea0a",
      ] -> null
      # (8 unchanged attributes hidden)

    - capacity_reservation_specification {
        - capacity_reservation_preference = "open" -> null
      }

    - cpu_options {
        - core_count            = 1 -> null
        - threads_per_core      = 2 -> null
          # (2 unchanged attributes hidden)
      }
```


Windows PowerShell
}
- credit_specification {
- cpu_credits = "unlimited" -> null
}
- enclave_options {
- enabled = false -> null
}
- maintenance_options {
- auto_recovery = "default" -> null
}
- metadata_options {
- http_endpoint = "enabled" -> null
- http_protocol_ipv6 = "disabled" -> null
- http_put_response_hop_limit = 1 -> null
- http_tokens = "optional" -> null
- instance_metadata_tags = "disabled" -> null
}
- primary_network_interface {
- delete_on_termination = true -> null
- network_interface_id = "eni-012741dbd3bfef9ba" -> null
}
- private_dns_name_options {
- enable_resource_name_dns_a_record = false -> null
- enable_resource_name_dns_aaaa_record = false -> null
- hostname_type = "ip-name" -> null
}
- root_block_device {
- delete_on_termination = true -> null
- device_name = "/dev/sda1" -> null
- encrypted = false -> null
- iops = 100 -> null
- tags = {} -> null
- tags_all = {} -> null
- throughput = 0 -> null


Windows PowerShell
- private_dns_name_options {
- enable_resource_name_dns_a_record = false -> null
- enable_resource_name_dns_aaaa_record = false -> null
- hostname_type = "ip-name" -> null
}
- root_block_device {
- delete_on_termination = true -> null
- device_name = "/dev/sda1" -> null
- encrypted = false -> null
- iops = 100 -> null
- tags = {} -> null
- tags_all = {} -> null
- throughput = 0 -> null
- volume_id = "vol-0f27164e2f72bddb7" -> null
- volume_size = 8 -> null
- volume_type = "gp2" -> null
(1 unchanged attribute hidden)
}
}
Plan: 0 to add, 0 to change, 1 to destroy.
Changes to Outputs:
- instance_id = "i-086979683bb1e560b" -> null
- instance_type = "t3.micro" -> null
- public_ip = "15.135.79.75" -> null
Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.
Enter a value: yes
aws_instance.student_vm: Destroying... [id=i-086979683bb1e560b]
aws_instance.student_vm: Still destroying... [id=i-086979683bb1e560b, 00m10s elapsed]
aws_instance.student_vm: Still destroying... [id=i-086979683bb1e560b, 00m20s elapsed]
aws_instance.student_vm: Destruction complete after 21s
Destroy complete! Resources: 1 destroyed.
PS C:\Terraform>


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Terraform-Stu...
i-086979683bb1e560b
Terminated
t3.micro
ap-southeast-2a
All instances
Instance alarm states
In alarm
None Show instances in alarm
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard
Health events affecting EC2 instances
Open events
None Show instances
Upcoming events
None Show instances
View AWS Health Dashboard
CloudShell
Agent Toolkit for AWS
Feedback
Language
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Privacy
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Cookie preferences

6. Real-World Technical Challenges & Solutions

Sr No.	Error / Issue Encountered	Root Cause	Engineering Solution Implemented
1	403 UnauthorizedOperation on ec2:DescribeImages	Organization Service Control Policy (SCP: p-81tb9lnn) blocked AMI search queries.	Replaced data "aws_ami" block with dynamic Systems Manager (SSM) Parameter Store lookup (aws_ssm_parameter).
2	AccessDeniedException on ssm:GetParameter	User Experiment7 lacked permissions to read SSM paths.	Attached managed IAM policy AmazonSSMReadOnlyAccess to IAM user Experiment7.
3	403 Explicit Deny on ec2:RunInstances	SCP enforced mandatory tagging on both EC2 instances and root EBS volumes.	Configured both tags and volume_tags blocks (Environment="Dev", Project="Experiment7") in HCL code.
4	InvalidParameterCombination: Not eligible for Free Tier	Sandbox environment enforced t3.micro instance type over t2.micro.	Updated instance_type = "t3.micro" in main.tf.

5	InvalidAMIID.Malformed	Hardcoded AMI ID mismatch across AWS regions.	Bound AMI lookup dynamically to active region (ap-southeast-2) via SSM Parameter Store.

7. Summary of Provisioned Infrastructure Attributes

Parameter Name	Deployed Value	Source / Verification Method
AWS Account ID	702561770822	aws sts get-caller-identity
IAM User	Experiment7	AWS IAM Console / CLI
AWS Region	ap-southeast-2 (Sydney)	HCL Provider Configuration
Instance Name Tag	Terraform-Student-VM	main.tf HCL Block
Instance ID	i-086979683bb1e560b	terraform output / AWS Console
Instance Type	t3.micro	AWS EC2 Instance Details
Public IPv4 Address	15.135.79.75	terraform output / AWS Console
Private IPv4 Address	172.31.7.17	AWS EC2 Instance Details
VPC ID	vpc-0e56bf12ccd240a63	AWS VPC Console / data.aws_vpc
Subnet ID	subnet-0ecc646eb242b088a	AWS Subnet Console / data.aws_subnets

Key Pair	vmb-key	AWS EC2 Key Pairs / main.tf
Lifecycle Final State	Terminated	terraform destroy / AWS Console

8. Conclusion

Learning Outcomes Achieved

Through this experiment, hands-on mastery was acquired in defining, provisioning, managing, and deprovisioning AWS cloud infrastructure using HashiCorp Terraform as an Infrastructure as Code (IaC) tool. The experiment successfully satisfied **Lab Outcome 4 (LO4)** by executing automated resource management workflows on cloud infrastructure.

Applications of the Studied Technique in Industry

- **Automated & Repeatable Deployments:** Eliminates manual human error in cloud console management by declaring infrastructure state in version-controlled repositories.
- **Multi-Cloud Architecture:** Terraform provides a unified framework to provision resources across AWS, Azure, GCP, and Kubernetes clusters using declarative HCL scripts.
- **Policy Compliance & Governance:** Seamlessly integrates with enterprise Service Control Policies (SCPs) and security frameworks to enforce automated tagging, network isolation, and access controls.

Engineering Relevance

Infrastructure as Code forms the foundational backbone of modern DevOps paradigms, Site Reliability Engineering (SRE), and CI/CD pipelines. Programmatically controlling infrastructure lifecycle allows engineering teams to rapidly create ephemeral testing environments, scale compute capacity dynamically, maintain exact environment parity across staging and production, and implement disaster recovery protocols.